

# Snake Game — Design Document

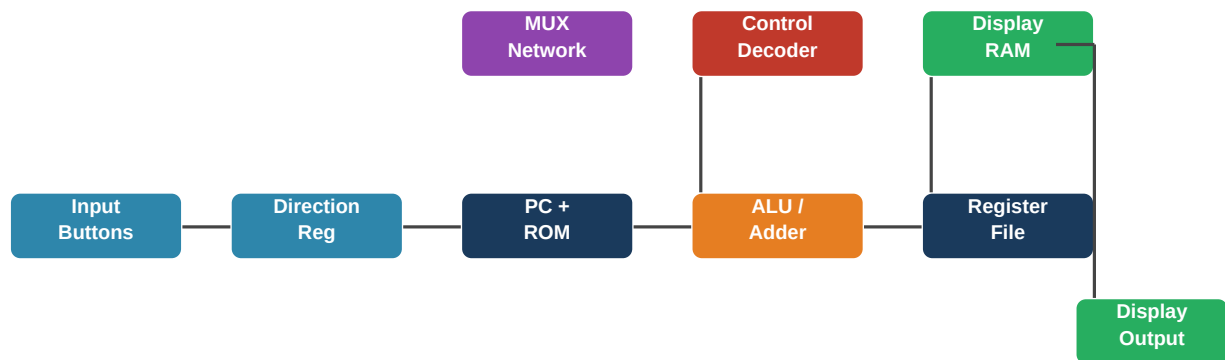
Logisim Evolution Digital Circuit Implementation

## 1. Purpose and Scope

This design document describes the hardware architecture of the Logisim Snake Game. It covers circuit block descriptions, signal flows, memory maps, opcode formats, and timing diagrams. It serves as a reference for understanding, modifying, or extending the design.

## 2. Block Diagram

High-level data flow between major circuit blocks:



## 3. Component Specifications

### 3.1 Program Counter (PC)

The PC is a 7-bit register that increments by 1 on each rising clock edge. It resets to 0x00 on system reset. The HALT instruction forces the PC into a hold state by gating the clock enable signal.

- Width: 7 bits (addresses 0x00 – 0x7F)
- Reset: synchronous, active-high
- Hold: driven by HALT control signal via AND gate on CE

### 3.2 Instruction ROM

Contents observed in simulation:

| Address | Value   | Decoded Meaning                 |
|---------|---------|---------------------------------|
| 0x080   | 0bc7c01 | Jump-and-link: game loop branch |
| 0x081   | 0440056 | ALU add: position increment     |
| 0x082   | 0000000 | NOP                             |
| 0x083   | 0000000 | NOP / Game Over state           |

### 3.3 Register File

The register file contains 8 general-purpose 7-bit registers plus dedicated registers for game state. Registers are read combinatorially and written synchronously.

| Register   | Width | Purpose                                        |
|------------|-------|------------------------------------------------|
| R0 (X)     | 7-bit | Snake head X coordinate                        |
| R1 (Y)     | 7-bit | Snake head Y coordinate                        |
| R2 (DIR)   | 2-bit | Current direction (00=UP, 01=DN, 10=LT, 11=RT) |
| R3 (LEN)   | 7-bit | Current snake length                           |
| R4 (FX)    | 7-bit | Food X coordinate                              |
| R5 (FY)    | 7-bit | Food Y coordinate                              |
| R6 (SCORE) | 7-bit | Current score                                  |
| R7 (FLAGS) | 4-bit | Status flags: Z, C, HALT, FOOD                 |

### 3.4 ALU

The ALU supports addition (position update) and comparison (boundary / collision check). Carry-in is provided from the c\_in register; carry-out is stored to c\_out for branching.

- ADD:  $A + B + C_{in}$
- SUB:  $A - B$  (boundary check)
- CMP: sets Z flag when  $A == B$

### 3.5 MUX Network

Multiple 2:1 and 4:1 multiplexers select data sources depending on the current instruction. The sel control bus (driven by the decoder) determines which inputs are forwarded to the ALU, register

write bus, and display RAM address lines.

### 3.6 Control Decoder

| Opcode | Mnemonic | Action                                 |
|--------|----------|----------------------------------------|
| 000    | NOP      | No operation; PC increments normally   |
| 001    | STORE    | Write head position to display RAM     |
| 010    | JMPZ     | Jump to target if Z flag = 1           |
| 011    | BPNT     | Branch if food coordinate matches head |
| 100    | HALT     | Set HALT flag; freeze PC               |

## 4. Memory Map

Position RAM (str/sel/ld/clr bus, as seen in simulation):

| RAM Address | str | sel | ld | clr | Description             |
|-------------|-----|-----|----|-----|-------------------------|
| 000         | 00F | 000 | —  | —   | Head X,Y = (0x0F, 0x00) |
| 002         | 00F | 000 | —  | —   | Segment 1 X,Y           |
| 004         | 00F | 001 | —  | —   | Segment 2 X,Y+1         |
| 006         | 000 | 000 | —  | —   | Tail / empty            |

## 5. Signal Timing

All registers are positive-edge triggered. One game tick = 1 clock cycle (in simulation this is manually stepped or run at a low Hz rate). The critical path runs: ROM output → MUX → ALU → Register write, which must settle before the next rising edge.

- tsetup (register): must be met by all datapath outputs before CLK↑
- thold: short — flip-flops are standard D-type
- Display update: occurs on same CLK edge as register write (combinational decode)

## 6. External Signal Interface

| Signal | Direction | Width | Description       |
|--------|-----------|-------|-------------------|
| CLK    | In        | 1     | Master clock      |
| RST    | In        | 1     | Synchronous reset |

|            |     |   |                                  |
|------------|-----|---|----------------------------------|
| BTN[3:0]   | In  | 4 | Up/Down/Left/Right buttons       |
| PIXEL_OUT  | Out | 1 | Current pixel value for display  |
| ADDR[5:0]  | Out | 6 | Display address for pixel scan   |
| HALT_LED   | Out | 1 | Game-over indicator              |
| SCORE[6:0] | Out | 7 | Current score (7-segment or raw) |